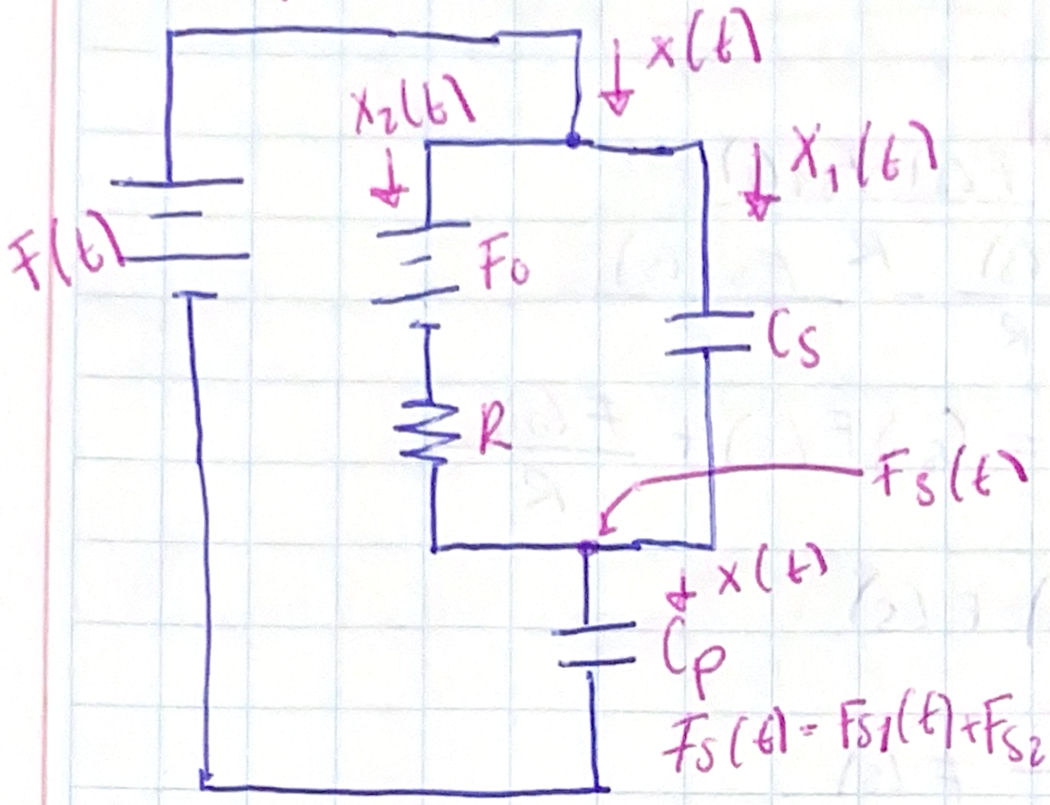
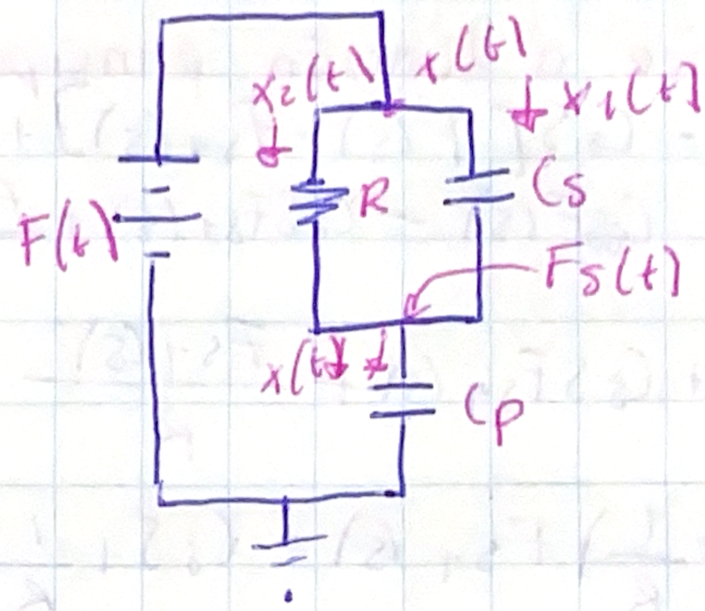
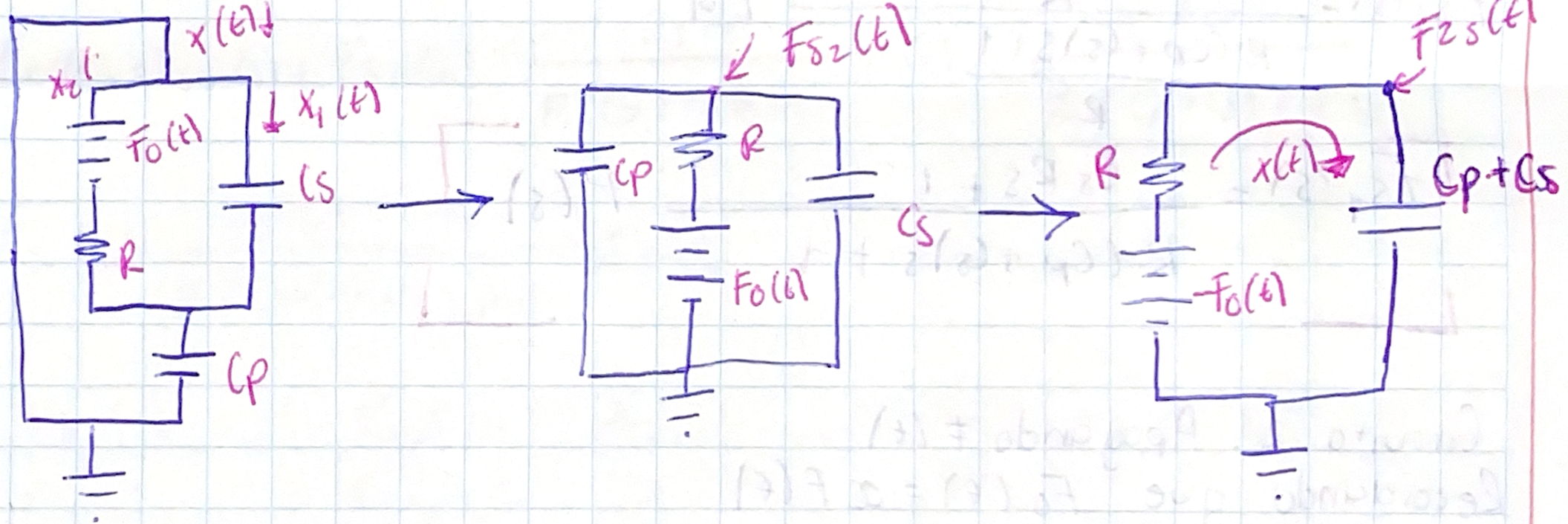
# 5.5. Práctica: Modelado y control del sistema musculoesquelético

$x(t) = x_1(t) + x_2(t)$




## Análisis matemático

Se aplica el principio de superposición

Circuito 1. Apagando $F_0(t)$

$$x(t) = x_1(t) + x_2(t)$$

$$x(t) = C_p \frac{dF_{s1}(t)}{dt}$$

$$x_1(t) = C_s \frac{d[F(t) - F_{s1}(t)]}{dt}$$

$$x_2(t) = \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p \frac{dF_{s1}(t)}{dt} = C_s \frac{d[F(t)-F_{s1}(t)]}{dt} + \frac{F(t)-F_{s1}(t)}{R}$$

Aplicar Laplace a función principal

$$C_p s F_{s1}(s) = C_s S[F(s)-F_{s1}(s)] + \frac{F(s)-F_{s1}(s)}{R}$$

$$C_p S F_{s1}(s) = C_s S F(s) - C_s S F_{s1}(s) + \frac{F(s)}{R} - \frac{R}{R} \frac{F_{s1}(s)}{R}$$

$$C_p S F_{s1}(s) + C_s S F_{s1}(s) + \frac{F_{s1}(s)}{R} = C_s S F(s) + \frac{F(s)}{R}$$

$$\left(C_p s + C_s S + \frac{1}{R}\right) F_{s1}(s) = \left(C_s S + \frac{1}{R}\right) F(s)$$

$$\frac{R(C_p + C_s)s + 1}{R} F_{s1}(s) = \frac{C_s R s + 1}{R} F(s)$$

$$F_{s1}(s) = \frac{\frac{C_s R s + 1}{R}}{\frac{R(C_p + C_s)s + 1}{R}} F(s)$$

$$\boxed{F_{s1}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)}$$

Circuito 2. Apagando $F(t)$

Recordando que $F_0(t) = \alpha F(t)$

$$-\alpha F(t) = Rx(t) + \frac{1}{C_p + C_s} \int x(t)\, dt$$

$$F_{s2}(t) = \frac{1}{C_p + C_s} \int x(t)\, dt$$

$$-\alpha F(s) = R x(s) + \frac{x(s)}{(C_p + C_s)s}$$

$$F_{s2}(s) = \frac{x(s)}{(C_p + C_s)s}$$

Estrella

$$F(s) = -\frac{1}{\alpha}\left(R + \frac{1}{(C_p + C_s)s}\right) x(s) = -\frac{1}{\alpha}\,\frac{R(C_p+C_s)s + 1}{(C_p + C_s)s}\,x(s)$$

$$\frac{F_{s_2}(s)}{F(t)} = \frac{\dfrac{x(s)}{(C_p+C_s)s}}{-\dfrac{1}{\alpha}\,\dfrac{R(C_p+C_s)s+1}{(C_p+C_s)s}\,x(s)} = -\frac{\alpha}{R(C_p+C_s)s+1}$$

$$F_{s_2}(s) = -\frac{\alpha F(s)}{R(C_p+C_s)s+1}$$

## Función de transferencia

$$F_s(s) = F_{s_1}(s) + F_{s_2}(s) = \frac{C_s R s + 1}{R(C_p+C_s)s+1}\,F(s) - \frac{\alpha F(s)}{R(C_p+C_s)s+1}$$

$$F_s(s)\,\frac{C_s R s + 1 - \alpha}{R(C_p+C_s)s+1}\,F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{C_s R s + 1 - \alpha}{R(C_p+C_s)s+1}$$

- Error en estado estacionario:

$$e(s) = \lim_{s\to 0} s\,F(s)\left(1 - \frac{F_s(s)}{F(s)}\right) = s\,\frac{1}{s}\left(1 - \frac{RC_s s + 1 - \alpha}{(RC_p + RC_s)s+1}\right)$$

$$e(s) = 1 - \frac{1-\alpha}{1} = 1 - 1 + \alpha = \underline{\alpha}$$

- Estabilidad en lazo abierto:
$(RC_p + RC_s)s + 1 = 0$

$$\lambda = -\frac{1}{RC_p + RC_s} = \text{Sistema estable (Número real negativo)}$$

Estrella